

AI Customer Support Agent for @AppleSupport

Hiver SDE Intern Take-Home Technical Report | Evaluation-Focused AI Prototype

Executive Summary

Social media customer support on Twitter/X presents acute challenges: communication is real-time, publicly visible, and constrained to 280 characters. For a brand of Apple's scale, an AI agent must not only provide high-accuracy troubleshooting; it must earn organizational trust by knowing **when not to answer**. This report documents an evaluation-focused AI Support Agent prototype targeting @AppleSupport, evaluated across a 200-sample hand-labelled holdout Golden Evaluation Set. Evaluating against two baselines (Trivial Majority-Rule and Classical Statistical ML), the agent achieves **69.5% out-of-sample Intent Accuracy** across a 7-class domain taxonomy, an **Escalation Recall of 63.8%** (catching human escalations vs. 0–3.5% for baselines), **79.5% official Apple link validity**, and **100% Twitter length compliance**. Reproduction runs completely offline on CPU in **6.2 seconds** with zero data leakage.

1. Problem Framing & Scope Boundaries

What 'Good' Means for @AppleSupport:

- **Safety and Privacy First:** Public social threads must never handle account credentials, passwords, 2FA codes, or billing card details. 'Good' means routing to private authenticated channels (DM or iforgot.apple.com / reportaproblem.apple.com).
- **Definitive Actionability:** Providing concrete, verifiable navigation paths (Settings > Battery > Battery Health) and canonical knowledge base links rather than generic advice.
- **Apple Brand Persona:** Calm, welcoming, professional, and empathetic tone within Twitter's 280-character budget.

What We Chose NOT to Build (Explicit Non-Goals):

- *No Automated Financial Payouts:* All refunds and subscription cancellations require authenticated human review; automated bots executing refunds on public social media create catastrophic fraud exposure.
- *No In-Chat Password/Credential Resets:* Account lockouts are strictly routed to official Apple recovery infrastructure.
- *No Machine-Translated Technical Troubleshooting:* Non-English queries (Spanish, French) are recognized and routed to native-language queues to avoid dangerous terminology errors in recovery steps.
- *No Speculative Hardware Cost Estimation:* Physical component damage is routed to Genius Bar scheduling.

2. Empirical Results vs. Two Baselines

We benchmarked three systems on the 200 hand-labelled Golden Evaluation Set using a strictly thread-disjoint training split (zero conversation ID overlap):

Metric Dimension	Trivial Baseline (Always Auto-Handle)	Simple Baseline (Naive Bayes + 1-NN)	Proposed AI Agent (RAG + Policy Engine)
Intent Accuracy (Out-of-Sample)	39.0%	58.0%	69.5%
Intent Macro F1	0.080	0.283	0.527
Escalation Accuracy	71.0%	71.5%	71.5%
Escalation Recall (Safety-Critical)	0.0%	3.5%	63.8%
Escalation Precision	0.0%	66.7%	50.7%
False Escalation Rate (Lower=Better)	0.0%	0.7%	25.4%
Twitter Char Compliance (<280)	100.0%	93.0%	100.0%
Official Domain Link Validity	0.0%	0.0%	79.5%
Intent-Link Relevance Rate	0.0%	0.0%	72.6%
Judge: Groundedness (1-5)	4.20	4.06	4.79
Judge: Brand Voice & Empathy (1-5)	5.00	4.31	4.58

Judge: Actionability (1-5)	3.40	4.31	4.81
Judge: Escalation Appropriateness (1-5)	3.95	3.98	4.34
Judge: Overall Quality Score (1-5)	4.14	4.16	4.63

The Deception of Accuracy in Baselines: Both Trivial and Simple baselines exhibit ~71% escalation accuracy simply by guessing AUTO_HANDLE for almost everything. However, their **Escalation Recall is 0.0% and 3.5%**. They fail to catch 96.5% to 100% of cases requiring human attention! The Proposed Agent catches 63.8% of escalations with 50.7% precision.

3. Failure Mode Analysis (Top 5 Failure Modes)

Failure Mode 1: Sarcasm and Idiomatic Frustration

- Query: 'It's been nearly two weeks and I've yet to get LTE on my Watch working. Think I paid a premium for a spec of red paint.'
- Classification: Model predicted DEVICE_SETUP_AND_USAGE (Auto-handle). Gold: CUSTOMER_FEEDBACK_COMPLAINT (Escalate).
- Hypothesis: Shallow embeddings latch onto 'LTE Watch working' without detecting the sarcastic 'spec of red paint'. Mitigation: Sarcasm detector sidecar and flagging any query mentioning unresolved timeframes > 7 days.

Failure Mode 2: Multi-Intent / Compound Inquiries

- Query: 'I updated to 11.0.2 and my phone is freezing. Also I was charged \$9.99 on my credit card without receipt!'
- Classification: Model predicted IOS_SOFTWARE_UPDATE (Auto-handle). Gold: APP_STORE_AND_BILLING (Escalate).
- Hypothesis: Single-label classifier prioritized the software update tokens that appeared first. Mitigation: Multi-label classification where safety-critical intents automatically dominate triage.

Failure Mode 3: Hardware Thermal Safety Ambiguity

- Query: 'My iPhone is burning up while charging.'
- Classification: Standard battery health auto-handle vs. lithium-ion safety hazard. Idiomatic heat ('burning up') can mean normal fast-charging warmth or a hazardous swelling battery. Mitigation: Clarifying safety prompt if heat terms appear.

Failure Mode 4: False Escalation on Standard FAQs with Negative Sentiment

- Query: 'I hate this update! Where is the shuffle button in Apple Music? It is impossible to find!'
- Classification: Escalated due to 'hate' and 'impossible'. Gold: Auto-handle. Mitigation: Decouple emotional sentiment from technical resolvability for clear UI navigation questions.

Failure Mode 5: Regional Dialects and Colloquial Slang

- Query: 'Awrite av got a problem with my iPhone padlock icon...'
- Classification: Subword segmenters encounter lower confidence on heavy phonetic slang, occasionally triggering low-confidence escalation. Mitigation: Phonetic slang normalization preprocessor.

4. 'What is Misleading About My Headline Number?'

A rigorous engineering evaluation requires confronting the blind spots of offline metrics:

1. **The Offline vs. Online Dynamic Gap:** Offline evaluation measures static, single-turn replies. An answer with a link receives a 5/5 score for actionability. In production, if the customer clicks the link, fails to resolve their problem, and tweets back angrily, the true First-Contact Resolution (FCR) is zero. Static evaluation cannot measure multi-turn resolution.
2. **The Asymmetric Cost of False Auto-Handles:** Reporting 71.5% escalation accuracy hides the fact that missing an account compromise or refund dispute costs ~\$100 in customer churn and liability, whereas an unnecessary human escalation costs ~\$4 in agent time. The operational cost curve is highly asymmetric.
3. **Domain Taxonomy Conditioning:** Achieving 69.5% accuracy across 7 classes reflects our chosen taxonomy boundary. If evaluated on open-vocabulary customer queries without pre-defined taxonomy bounds, performance would decline.
4. **LLM Judge Leniency Bias:** Automated rubrics have an inherent preference for grammatically polished text with official-sounding URLs, occasionally rating an authoritative-sounding outdated troubleshooting step too generously.
5. **Twitter Channel Selection Bias:** Public tweets over-represent tech-savvy users experiencing public software update bugs and under-represent complex hardware repairs handled via phone support.

5. Human-Judge Inter-Rater Reliability (N=200)

To establish trust in our automated evaluator, we measured alignment between judge ratings and human annotations across all 200 items:

- **Pearson Correlation:** $r = 0.343$ on Overall Score and $r = 0.392$ on Escalation Appropriateness, indicating moderate linear alignment.
- **Mean Absolute Error:** **0.290 points** on the raw 1.0–5.0 scale, with **90.0% of all ratings within 0.5 points** of the human ground truth.
- **Cohen's Kappa:** $\kappa = 0.222$ on binned quality tiers, confirming agreement above random chance without artificial score inflation.

6. What We Would Do Next with One More Week

1. **Tri-State Copilot Routing:** Implement High Confidence (>0.85) -> Auto-Reply; Medium Confidence (0.55–0.85) -> One-Click Draft in Hiver inbox for human agent review; Low Confidence -> Direct Escalation.
2. **Stateful Multi-Turn Dialog Trees:** Track user conversation state and ask structured disambiguating questions when symptoms are vague.
3. **Mock CRM Tool Calling:** Connect agent to live Apple System Status API endpoints and AppleCare warranty entitlement lookups before generating replies.
4. **Active Learning Loop:** Automatically route low-confidence customer queries to human reviewers daily to continuously expand the golden benchmark.